

DIGITAL *adj.* – Fr.: *numérique*; Sp.: *digital*.

Definition: relating to the electronic storing and processing of information using a fixed set of digits (either a '0' or '1', designated for positive and non-positive signals), but also more generally denoting activity related to the use of computer technology. Digital relates to multiple aspects of museum work. Museums *use* digital technology as a tool – the software and hardware, platforms and media technologies, that enable the organisation to meet its mission. Likewise, museums *manage* digital processes – the systems and strategies, policies and project management methods around an organisation's digital operation. However, museums also *create* digital things – including computer-based records, in-gallery media content, online podcasts, publications and social media interactions. And yet, museums also *understand* 'digital' as a concept and culture – a societal context in which they operate, a set of visitor expectations around technology, and a subject within their curation.

The adoption of, and adaption to, digital technology has been referred to in different times, and in different professional and cultural contexts as: Museum Computing; Museum Informatics, Digital Heritage; Digital Cultural Heritage; Museum Technology; and even (latterly) simply as #musetech.

Related terms: authenticity, display, documentation, information, interactivity, management, media, visitor experience

The (modernist) advent of computer-based technology

The first major influences of digital technology on the museum were largely related to collections management. During the last quarter of the twentieth-century, digital technology was closely implicated in the sector's move to more uniform systems of museum documentation. At that time, during the 1960s and 1970s, as the museum sector sought a professionalisation of its approach to managing its ever-growing collections, a new generation of computers were becoming more widely available and affordable to many organisations, with the promise of the efficient programmable processing of large amounts of structured data. For museums, this confluence (of looking to systematize their approaches at a concurrent step-change moment in the history of computing), marked not only the new era of 'automation' of manual documentation processes, but also the creation of enduring sets of standards (of 'codes', of 'fields', of 'terminology control', of 'standard units of information'), through which digital thought would continue to leave its mark for decades to follow. It was this digital thinking and the creation of a core 'data model' for museum records, that guided the fundamental conceptualisation of collections management. The 'collections database' stands, consequently, as the synecdoche of late twentieth-century museum practice.

In the age of artificial intelligence (AI) technology and the distributive network and logic of the Web, new sets of challenges and questions now surround these museum collections processes and data. Today, for example, museums ask what a genuinely object-orientated collections management system would look like – one that more closely reflects a modern museological mindset, one that starts with the unique 'needs' and 'value' of each individual object, rather than with the system's orthodox *a priori* categories and uniform questions. Similarly, today's challenge is understanding the consequence of applying the algorithms of 'machine learning' (computers detecting patterns in collection datasets) and 'deep learning' (computers making decisions based on these patterns) to the massification of data now in museums, and understanding how this might change not just the role of the curator and the

collections manager, but also what a museum collection, in the world, is for (Murphy & Villaespesa, 2020). And how these developments (of data modelling and data processing) might relate to a data ecology that is not only locally connected but globally networked, where museum collections can be inter-linked and aggregated. And so, just with the advent of documentation standards, where digital thinking and digital practice shaped the logic of the new systems, so today computing technology again provides both the provocation and the promise of solutions to collections management.

The (postmodern) disruption of digital technology in the museum

Beyond collections, the influence of digital technology on in-gallery design and experience has been equally profound. Since its arrival in exhibition spaces (from the 1980s onwards, but more widespread from the 1990s), digitally-mediated communication has evolved into an innate function of museum display (Samis & Michaelson, 2016). Initially, owing to their (then) size, their noise and their cumbersome appearance, the presence of these early gallery computers was largely anomalous to both the aesthetic and the ambition of most exhibitions. Typically, these 'kiosk' interactives of this nascent period of in-gallery interaction were physically and conceptually separated within the visitor's experience – more disruptive than instructive. However, as in-house and out-sourced design confidence grew, as costs reduced, visitor expectations rose, hardware became more versatile and diverse (in size, shape and orientation), and software more adaptive and adaptable (in interface and function), so the possibilities for integrating digital technology into the space of the gallery increased. Consequently, from being dislocated, in-gallery digital technology became integrated (actively part of), and then in time embedded within (and seamless to) the concept and experience of the exhibition.

Gallery design has been affected by the advent of projection-based display (enabling large-scale and immersive spectacle); of movement sensing and gesture recognition (implicating the actions of the visitor); of ambient and directional sound technology (helping to sequence the experience, and at times preserve the personal experience in the public space of the gallery); of forms of haptic feedback (re-negotiating what can and what cannot be touched, what is and what is not out of reach, present or not present); and of modes of media reality (extending the illusory traditions of museum display through the digital compositions of 'augmented reality' and the theatrical ensembles of 'mixed reality') (Peng, 2019). These technological developments, and others like them, embraced by the museum, have together rendered the gallery space not only a media rich space, but a 'mediascape' (Kidd, 2014) in which the technology can be both pervasive and ambient – potentially everywhere, but also potentially invisible. Just as other display technologies (lockable cabinets, emblematic labelling, uniformly transparent glass, reliable gas lighting, environmentally stable casing, vinyl cutting) were all once their period's 'new technology', but in time became orthodoxy, so the once novelty of digital-mediated communication has today become a function of what 'exhibition' is.

Equally defining for this new media ensemble, since the turn of the century, has been the rise mobile devices within the museum. Building on a long (pre-digital) tradition of locative guide technology (Tallon & Walker, 2008), the modern convention of visitors bringing their own mobile technology has presented the museum with a new set of potentials: from alternative forms of interaction and exchange between visitors (and non-visitors); to creative forms of play and gamification of the visit; and more flexible and accessible forms of user experience (Baggesen, 2019). In this way, the visitor's ever-present mobile device (as a tool for navigating, for making, for communicating, for recording, for publishing) symbolizes and enables a trajectory towards ever-more participatory and personalized gallery experiences.

Similar to mobile media, the networked technology of the internet generally, and the Web specifically, remain catalytic for the museum. Multiple and varied channels of social media content generate alternative norms for museum experience, beyond the physicality of the traditional curated visit experience (Drotner & Schrøder, 2013). The distributiveness of open online content, 'virtual collections', and digital aggregations of objects, constructively question assumptions not just around what a collection is, but where it is and who owns it. Likewise, the collective intelligence of user-generated content offers a new locus, voice, and identity for expertise, beyond curatorial interpretation and authority (Ridge, 2016). And the informational and media-based nature of digital content, supports the museum in its thinking beyond the primacy of material things. And so, confidently at odds with the centralizing institutions that have preceded it, the twenty-first century museum is instead increasingly self-assured with distributing collections, to distributed users, through a model of distributive expertise.

The (metamodernist) maturity of digital practice in the museum

Underpinning these uses of digital tools and this creation of digital content, is the museum's maturity in managing digital processes. A digitally mature museum (a 'postdigital museum') is one in which computer technology, rather than be approached as something new, unfamiliar or ancillary, is instead managed normatively and confidently across the organisation (Parry, 2013): the museum's visioning and articulation of itself prioritising digital; the organisation's leadership and leaderful behaviours activating digital; the museum's systems and business processes enabling digital; its institutional culture supporting digital; and – crucially – its staff ultimately driving digital.

This digital maturity is also manifest in how the museum uses technology in a 'purposeful' way, namely: that the museum recognizes its social purpose (and not just technocratic efficiency) as the principal motivation for pursuing organisational digital transformation; and that the museum's socially purposeful work not only leverages the consequences of the organisation's own digital transformation, but also acknowledges that the society in which the museum intends to be purposeful is itself digitally transformed. Combining its ambitions of being both 'digitally transformed' and 'socially purposeful', this is the 'digitally purposeful' museum.

These forms of digitalisation are, however, not inevitable, nor indeed are they uniform for all museums around the world. Museums, after all, vary in their size and resource, business models and governance, subject and purpose. Consequently, each organisation's approach to adopting, and adapting to, computer technology is determined by its own context and priorities. Each museum defines its own relationship with digital technology based on its current levels of digital confidence, the surrounding policy expectation and resource opportunities, as well as its publics expectations around technology (Malde et al, 2019). Working holistically (across the organization) and contextually (on its own terms), each museum finds the combination of digital skills it needs within its own circumstance to build its digital confidence: its competency (its actions, and what it can do *with* digital technology); its capability (its intentions, and what it can achieve *through* digital technology); and its literacy (its reflections, and what it understands *about* digital technology).

Over the last half-century, museums have experienced a series of technology-related transitions: from system-orientated documentation to object-orientated responses; from single-site visitation to cross-platform provision; from centring the primacy of the material object to the staging of mixed-reality experiences; and (imminently) from an era of binary

computing, to an emerging era of quantum computing. As it continues to make these transitions, the museum is confronted with a set of beacon priorities and radical questions related to its work with digital technology. Ethical questions – acknowledging the museum's duty of care in digital space, its non-negotiable responsibilities for universal and accessible design, and the legal and rights implications of contributing to an open data, networked environment. Inclusive questions – confronting the prejudice that can be fossilized and so perpetuated within a museum's data models, and what it means to de-colonise museum data. As well as affective questions – recognising the emotionality not just of being confronted with digital technology as a visitor, but of being part of digital transformation in the museum as a worker, of leading digital change, particularly at times of institutional and individual precarity.

At its advent, the principal task for museum computing was on *how to order* the new efficient databases of modernity; the 'computerised museum' was a new and idealized frame for an automated and informational age. Subsequently, the main critique for museums and technology turned to *what was real* and what could be trusted within the new playful online provision of postmodernity; there, the 'virtual museum' became an ironic de-framing for a more questioning and skeptical age. And yet, today (whilst these former traditions still endure) attention moves to new priorities on *who has agency* within the activated networks and platforms of metamodernity; here, the 'online museum' stands as a hopeful act of re-mixing, for what can be a more resolved and inclusive age. As such, the story of museums and digital technology can, across these cultural conditions, be read as the story of the museum in society: from a catalogue to be built, to a collage to be questioned, to a community in which now to belong.

Ross Parry

References

- Baggesen, R. H. (2019). Mobile media, mobility & mobilization in the current museum field, in Drotner, K., Dziekan, V., Parry, R. & Schröder, K. C. (Eds), *The Routledge Handbook of Museums, Media and Communication* (pp. 115-127). London & New York: Routledge.
- Drotner, K. & Schröder, K. C. (Eds.) (2013). *Museum Communication and Social Media: The Connected Museum*. New York & London: Routledge.
- Kidd, J. (2014). *Museums in the New Mediascape: Transmedia, Participation, Ethics*. London & New York: Routledge.
- Malde, S., Kennedy, A. & Parry, R. (2019). *Understanding the Digital Skills & Literacies of UK Museum People*. Phase Two Report, One by One. Leicester: University of Leicester.
- Murphy, O. & Villaespesa, E. (2020). *AI: A Museum Planning Toolkit*. London: Goldsmiths, University of London.
- Parry, R. (2013). The end of the beginning: normativity in the postdigital museum, *Museum Worlds*, 1, 24-39.
- Peng, J. (2019). How does this exhibition make you feel? Measuring sensory and emotional experience of in-gallery digital technology with GSR devices, in Cherry, R. and Proctor, N. (eds.) *Museums and the Web 2019* (pp. 247-260). Silver Spring, MD: Museums and the Web.
- Ridge, M. (Ed.) (2016). *Crowdsourcing our Cultural Heritage*. Farnham & Burlington VT: Ashgate.
- Samis, P. & Michaelson, M. (2016). *Creating the Visitor-centered Museum*. New York & London: Routledge.
- Tallon, L. & Walker, K. (2004). *Digital Technologies and The Museum Experience: Handheld Guides and Other Media*. Lanham, Md & Plymouth: AltaMira Press.